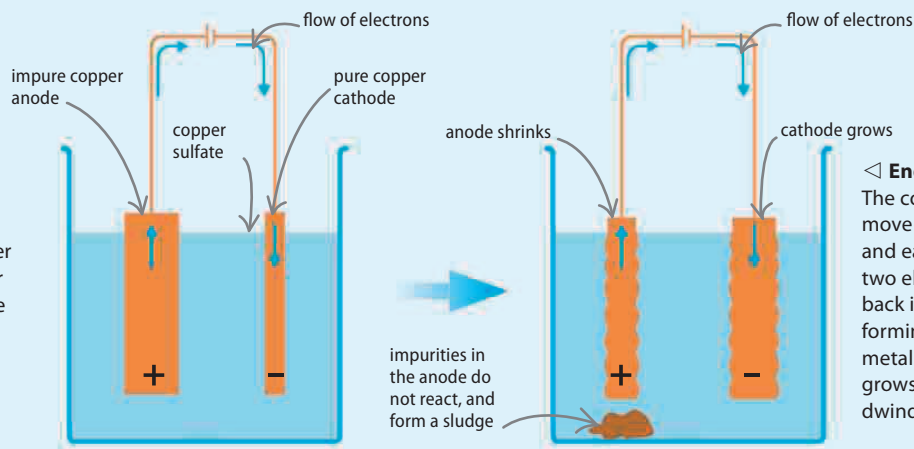


Purifying metals

Electrochemistry can be used to remove impurities from a metal and make an extremely pure sample. The piece of impure metal is used as the anode. A pure sliver of the same metal is the cathode. When the current is switched on, the metal ions in the impure metal leave the anode and dissolve in the electrolyte, a copper sulfate solution. Here they are free to move to the cathode, where they receive electrons and turn back into metal atoms.

Start point

The current turns each copper atom in the impure copper anode into a copper ion (Cu^{2+}). They leave the anode and dissolve into the copper sulfate electrolyte.



End point

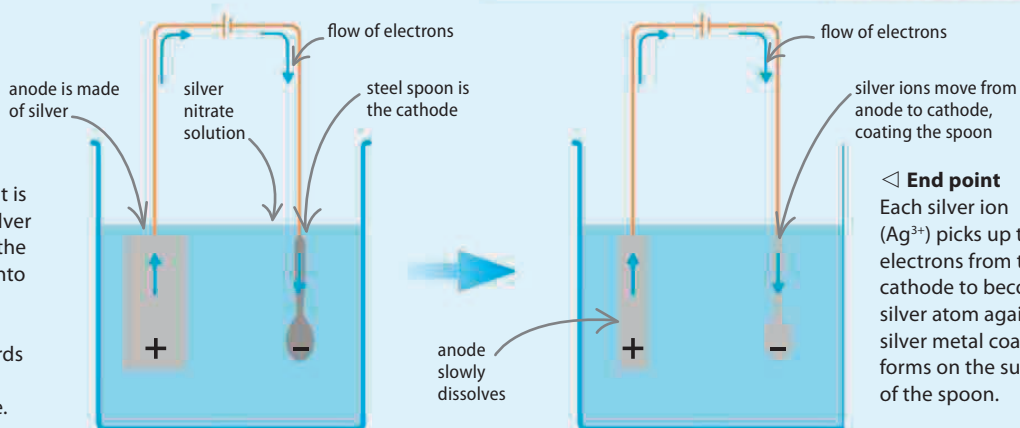
The copper ions move to the cathode and each ion receives two electrons to turn back into an atom (Cu), forming a layer of pure metal. So the cathode grows as the anode dwindles away.

Electroplating

A thin layer of precious metal can be added to a less expensive metal object using a process called electroplating. A piece of precious metal, such as gold or silver, is used as the anode. The item to be plated forms the cathode. The electrolyte also contains ions of the precious metal. The current makes the anode gradually dissolve and the precious metal ions transfer to the cathode, where they coat the object.

Start point

When the current is turned on, the silver ions (Ag^{3+}) from the anode dissolve into the silver nitrate electrolyte. Then they move towards the negatively charged cathode.



End point

Each silver ion (Ag^{3+}) picks up three electrons from the cathode to become a silver atom again. A silver metal coating forms on the surface of the spoon.

REAL WORLD

Galvanization

Electroplating can be used to coat and protect steel with a layer of zinc to make galvanized steel. This zinc-plated steel is more resistant to corrosion than iron (the main constituent of steel).

